

Homework 5

Problem 1 -- M2 decides $A = \{0^{2^n} \mid n \geq 0\}$

(a) input: 0

q1 0

$\vdash \sqcup q2 \sqcup$

$\vdash \sqcup q_accept \sqcup$

Result: **ACCEPT**

(b) input: 00

q1 00

$\vdash \sqcup q2 0$

$\vdash \sqcup x q3 \sqcup$

$\vdash \sqcup x q5 \sqcup$

$\vdash \sqcup q5 x \sqcup$

$\vdash q5 \sqcup x \sqcup$

$\vdash \sqcup q2 x \sqcup$

$\vdash \sqcup x q2 \sqcup$

$\vdash \sqcup x q_accept \sqcup$

Result: **ACCEPT**

(c) input: 000

q1 000

$\vdash \sqcup q2\ 00$

$\vdash \sqcup x\ q3\ 0$

$\vdash \sqcup x\ 0\ q4\ \sqcup$

$\vdash \sqcup x\ 0\ x\ q3\ \sqcup$

$\vdash \sqcup x\ 0\ x\ q5\ \sqcup$

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-> q_reject (odd number of 0s -- not a power of 2)

Result: **REJECT**

Problem 2 -- M1 decides $B = \{w\#w \mid w \in \{0,1\}^*\}$

(a) input: 11

q1 11

$\vdash x q3 1$

$\vdash x 1 q3 \sqcup$

$\rightarrow q_reject$ (no # in input)

Result: **REJECT**

(b) input: 0#0

q1 0#0

$\vdash x q2 \#0$

$\vdash x \# q4 0$

$\vdash x \# x q6 \sqcup$

$\vdash x \# q6 x$

$\vdash x q6 \# x$

$\vdash q7 x \# x$

$\vdash x q1 \# x$

$\vdash x \# q8 x$

$\vdash x \# x q8 \sqcup$

$\vdash x \# x q_accept \sqcup$

Result: **ACCEPT**

(c) input: 0##0

q1 0##0

$\vdash x q2 \##0$

$\vdash x \# q4 \#0$

$\rightarrow q_reject$ (q4 encounters second # with no valid transition)

Result: **REJECT**

(d) input: 1#0

q1 1#0

$\vdash x q3 \#0$

$\vdash x \# q5 0$

-> q_reject (q5 expects 1 on right side, found 0)

Result: **REJECT**